



ITCS442

Relational Database Design

Project Phase I: Database Planning and Requirement Analysis

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Presented to

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Introduction

Project Title: Restaurant Inventory Management System

Group ID: 5

Group Members

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Declaration of AI Tool Usage

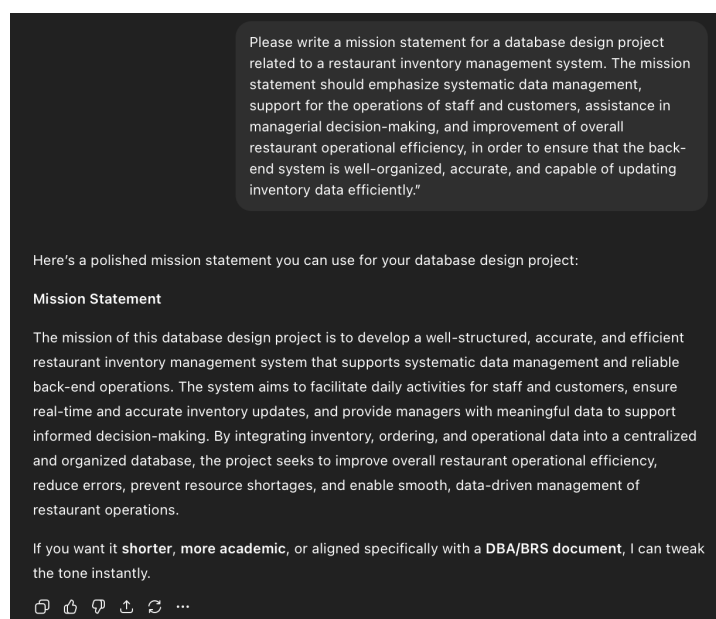
This project applies **AI Usage Level 2 (AI-assisted planning and structuring)**. Artificial intelligence tools were used only to support brainstorming, planning, and organizing ideas. All content was reviewed, refined, and finalized by the project members.

AI Tools Used: ChatGPT :

1st Attempt:

- **Prompts:**“Please write a mission statement for a database design project related to a restaurant inventory management system. The mission statement should emphasize systematic data management, support for the operations of staff and customers, assistance in managerial decision-making, and improvement of overall restaurant operational efficiency, in order to ensure that the back-end system is well-organized, accurate, and capable of updating inventory data efficiently.”

- **Date of Use:** 19 January 2026



AI Tools Used: ChatGPT :

2st Attempt:

● **Prompts:**

● **Date of Use:** 23 January 2026

AI Tools Used: ChatGPT :

3st Attempt:

● **Prompts:**

● **Date of Use:** 25 January 2026

Motivation

Due to the increasing demand for fast and convenient food services, restaurants must manage their ingredients efficiently to ensure smooth daily operations. Accurate control of ingredient availability and stock levels is essential to prevent shortages and reduce waste. However, many restaurants still rely on manual inventory recording, which makes it difficult to track ingredient usage and stock status effectively.

Manual inventory management increases the risk of errors, inconsistent data, and delayed updates. These issues can result in ingredient shortages, overstocking, and inefficient purchasing decisions, leading to higher operational costs and food waste.

Therefore, developing a Restaurant Inventory Management System is necessary to improve inventory accuracy and enable real-time stock monitoring. A centralized database system can reduce human errors, support better decision-making, and enhance overall operational efficiency in restaurant management.

As-Is Analysis

Current Restaurant Inventory Management System

Currently, the restaurant's inventory management process relies on manual operations using paper records and spreadsheets to manage ingredient and stock information. Data such as ingredient details, available quantities, supplier information, and stock update records are recorded separately and are not stored in a centralized system. This makes it difficult to accurately monitor inventory levels and maintain consistent, up-to-date information across the restaurant.

Problems in the Current System

Since inventory data is recorded manually, employees are required to physically count ingredients and enter stock quantities by hand. This process increases the likelihood of human error, data inconsistency, and delayed updates. In addition, the current system does not support automatic monitoring of minimum stock levels or alerts for low or out-of-stock ingredients, which often results in ingredient shortages or unnecessary over-ordering.

Furthermore, supplier information and purchase-related data are not managed in a structured manner, making it difficult to track price changes, ordering history, and supplier performance. These limitations reduce the efficiency of inventory planning and decision-making. Overall, the manual inventory management system leads to operational delays, increased waste, and reduced efficiency, negatively impacting the restaurant's overall performance.

1. Users

1.1 Manager

The manager uses the back-office system to review orders, menu information, and inventory status, including ingredient levels, in order to support planning, cost control, and overall restaurant operations.

1.2 Admin

The admin is responsible for managing and maintaining core system data, including customer, staff, menu, ingredient, and inventory information, as well as controlling system access and administrative functions to ensure data accuracy and consistency.

1.3 Staff

Staff members use the back-office system to process and update order status, record ingredient usage, and monitor inventory levels to support daily operations.

2. Core Entities

2.1 Manager

Description: Restaurant managers who have access to the back-office system to supervise operations, review orders, menu information, and inventory status, and update limited system data.

Data Entered: Manager ID, name, username, password, account creation date

2.2 Admin

Description: Administrators responsible for overall system management. Admins can manage all system data, control user access, and access system reports.

Data Entered: Admin ID, first name, last name, username, password, account creation date

2.3 Staff

Description: Restaurant employees who handle daily operational tasks such as managing orders, updating order status, and recording ingredient usage, with access limited to necessary operational data.

Data Entered: Staff ID, citizen ID, first name, last name, position, gender, employment record creation date

2.4 Order

Description: Customer orders received by the restaurant. Orders are used to record and track purchases and are linked to menu items in order to identify the ingredients used for each order and support inventory control.

Data Entered: Order ID, order date, total price, order status

2.5 Menu

Description: Food and beverage items offered by the restaurant. Menu information is used during order processing and to define the ingredients required for each menu item, supporting ingredient preparation and stock availability checks.

Data Entered: Menu ID, menu name, description, price, category

2.6 Ingredient

Description: Ingredients used in food preparation. This entity serves as a reference for tracking ingredient usage and managing food cost control.

Data Entered: Ingredient ID, ingredient name, unit of measurement, supplier, price per unit

2.7 Inventory

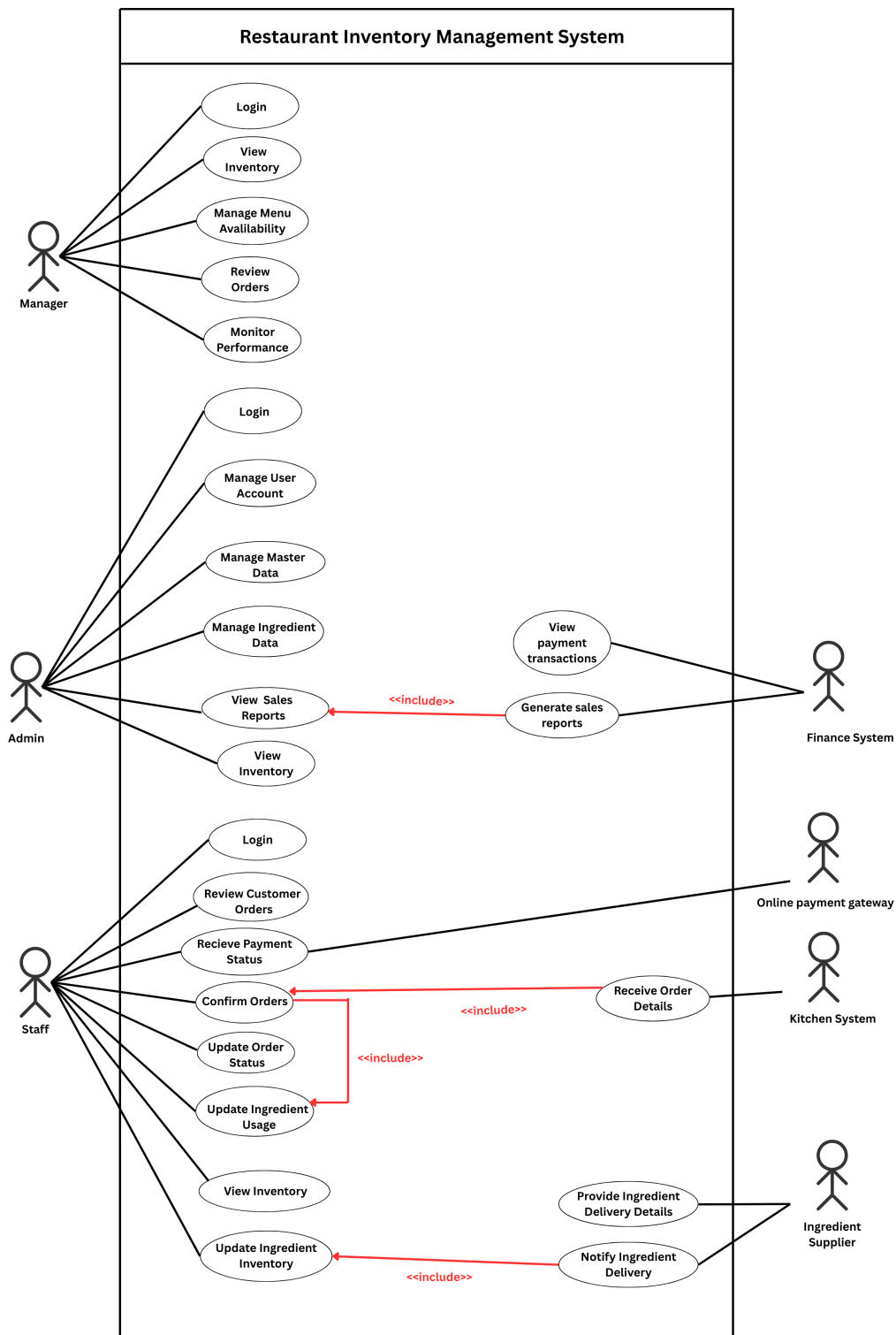
Description: The current stock levels of ingredients in the restaurant. It is used to monitor available quantities, define minimum stock levels, and support inventory control decisions.

Data Entered: Inventory ID, ingredient ID, quantity available, last updated date, minimum stock level

3. Key Business Rules

1. An order cannot be confirmed if required ingredient quantities are insufficient in inventory.
2. Each menu item must be associated with one or more ingredients.
3. Inventory quantities must be automatically updated after an order is confirmed.
4. Payment must be successfully confirmed before an order status can be changed to Completed.
5. Menu items marked as unavailable cannot be ordered.
6. Only authorized staff members are permitted to confirm orders and update order status.

Use Case Definition



https://www.canva.com/design/DAG_hyeHqtU/hQx-K_n4H4SiAiDbXgjedA/view?utm_content=DAG_hyeHqtU&utm_campaign=designshare&utm_medium=link2&utm_source=uniqueLinks&utm_id=hdb51884a25

Proposed Database Application

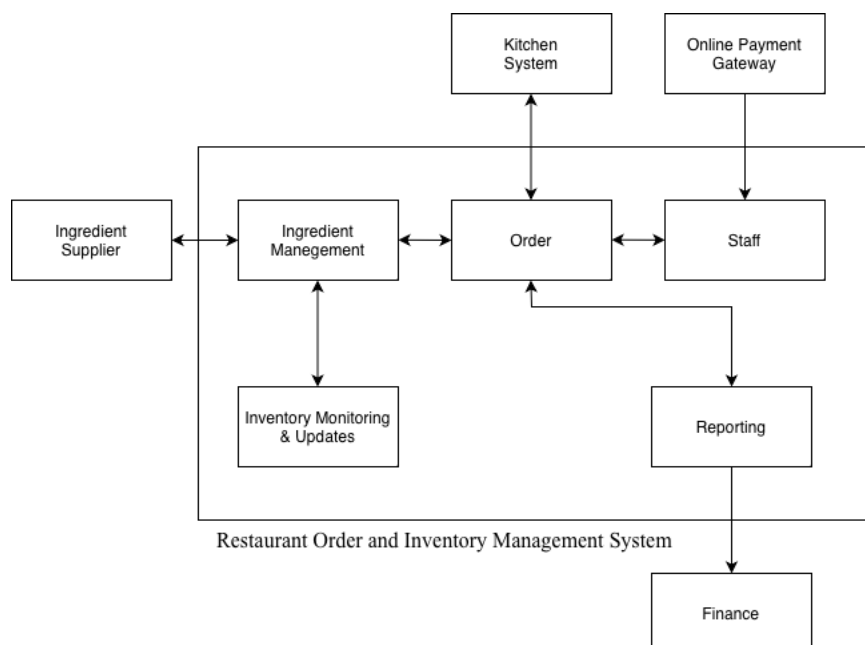
1. Mission Statement

The mission of this database design project is to create a structured, accurate, and efficient restaurant inventory management system that supports proper data handling and dependable back-end processes. The system is intended to assist staff and customers in their daily operations, maintain timely and precise inventory updates, and supply managers with valuable information to support sound decision-making. By consolidating inventory, ordering, and operational information into a centralized and well-organized database, the project aims to enhance overall restaurant performance, minimize operational errors, reduce the risk of stock shortages, and enable smooth, data-driven management of restaurant activities.

2. Mission Objectives

- To manage and integrate operational data to support accurate and timely decision-making for restaurant management.
- To support efficient creation and management of customer orders.
- To maintain accurate and up-to-date records of menu items and ingredients.
- To enable effective monitoring of operational data with controlled and consistent access.
- To ensure system scalability for future business growth.

3. Scope and Boundary



The system boundary includes order, ingredient management, inventory monitoring and updates, staff, and reporting. External entities outside the system boundary include ingredient suppliers, the kitchen system, the online payment gateway, and the finance system, which interact with the system only through data exchange. Moreover, staff manage

orders and monitor inventory levels, while payment status is received from the payment gateway. Ingredient management and inventory updating ensure accurate tracking and control of ingredient stock. Confirmed orders are sent to the kitchen system for food preparation. Completed transaction data is stored in the reporting module, and summarized information is shared with the finance system for financial analysis.

4. Mapping of Major User Views and Operations

Data	Access Type	Admin	Staff	Manager
Manager	Maintain	✓	✗	✗
	Query	✓	✗	✓
	Report	✓	✗	✗
Admin	Maintain	✓	✗	✗
	Query	✓	✗	✗
	Report	✓	✗	✗
Staff	Maintain	✓	✓	✗
	Query	✓	✓	✓
	Report	✓	✗	✗
Order	Maintain	✓	✓	✗
	Query	✓	✓	✓
	Report	✓	✗	✗
Menu	Maintain	✓	✓	✗
	Query	✓	✓	✓
	Report	✓	✗	✗
Ingredient	Maintain	✓	✓	✗
	Query	✓	✓	✓
	Report	✓	✗	✗
Inventory	Maintain	✓	✓	✗
	Query	✓	✓	✓
	Report	✓	✗	✗

Explanation of Data Access

This table defines the roles of each user group, and their data access needs in terms of query, maintain, and report operations. The access control is designed according to user responsibilities to ensure proper data management, data consistency, and system security. The access types are defined as follows.

- Maintain refers to the ability to create, update, or delete data.
- Query refers to the ability to view or search for information.
- Report refers to the ability to generate analytical or summary reports based on system data.

Explanation of User Access

1.Admin

- Has full control over the back-office system.
- Can create, update, and delete data across all system entities.

- Has exclusive access to reporting and analytical functions.
- Responsible for system configuration, user management, and data governance.
- Example: An admin can manage user accounts, maintain master data, and generate sales and inventory reports to support strategic decision-making.

2. Manager

- Has supervisory access to monitor and review operational data.
- Can view orders, menu information, and inventory status.
- Supports operational planning and decision-making based on system information.
- Does not have permission to perform data maintenance, system configuration, or generate administrative reports.
- Example: A manager can review daily orders, monitor ingredient stock levels, and evaluate menu performance to support planning and cost control.

3. Staff

- Handles day-to-day operational tasks within the system.
- Can manage orders, update order status, record ingredient usage, and update inventory quantities.
- Can view relevant operational data required for daily work.
- Does not have permission to generate reports or manage system configurations.
- Example: A staff member can confirm customer orders, and update inventory after order processing.

5. Users' Requirements Specification

Data Requirements

User / Entity	Data
Manager	• manager_id (PK) – Unique identifier for each manager • first_name – First name • last_name – Last name • username – Login username • password – Encrypted password • created_at – Account creation date
Admin	• admin_id (PK) – Unique identifier for each admin • first_name – First name • last_name – Last name • username – Login username • password – Encrypted password • created_at – Account creation date
Staff	• staff_id (PK) – Unique identifier for each staff • citizen_id – National identification number • first_name – First name • last_name – Last name • position – Job title (e.g., Chef, Waiter) • gender – Male / Female / Other • created_at – Employment record creation date

Order	● order_id (PK) – Unique identifier for each order ● order_date – Date and time of the order ● total_price – Total cost of the order ● order_status – Order status (Pending, Completed, Canceled)
Menu	● menu_id (PK) – Unique identifier for each menu item ● menu_name – Name of the dish ● description – Brief description of the menu item ● price – Selling price ● category – Menu category (e.g., Appetizer, Main Course, Dessert)
Ingredient	● ingredient_id (PK) – Unique identifier for each ingredient ● ingredient_name – Name of the ingredient ● unit – Unit of measurement (kg, liters, pieces, etc.) ● supplier – Supplier name ● price_per_unit – Cost per unit
Inventory	● inventory_id (PK) – Unique identifier for each inventory record ● ingredient_id (FK) – Reference to the ingredient ● quantity_available – Current stock quantity ● last_updated – Date and time of last inventory update ● minimum_stock_level – Minimum stock level before restocking

Transaction Requirements

User / Entity	Data Entry (Insert)	Data Update / Deletion	Data Queries
Manager	● Add new menu items ● Add new ingredient records	● Update menu availability ● Adjust menu prices	● View orders ● View inventory status ● Monitor sales performance
Admin	● Create admin, manager, and staff accounts ● Add menu items and ingredients	● Update or delete menu, ingredient, inventory, and user data	● Generate sales reports ● Generate inventory reports ● View system-wide data
Staff	● Confirm customer orders ● Record ingredient usage	● Update order status ● Update inventory quantities (e.g., Delete a staff member who resigned)	● Review customer orders ● View inventory information
Order	● Place a new order	● Update order status to “Completed” ● Cancel an order	● Retrieve orders placed by a specific customer ● Find all pending orders
Menu	● Add a new dish to the menu	● Update menu price (e.g., Change price from 120 to 140 Baht) ● Remove a discontinued dish	● List all menu items in a specific category (e.g., “Burger”) ● Find menu items below a specified price

Ingredient	● Add a new ingredient to the system (e.g., Add tomatoes supplied by FreshFarm)	● Update ingredient details (e.g., Change supplier name) ● Delete an expired ingredient	● Retrieve ingredient information ● Search ingredients by supplier
Inventory	● Create a new inventory record for an ingredient	● Update stock quantity after order confirmation ● Adjust inventory quantities during restocking	● List ingredients with stock below minimum level ● Retrieve current inventory status

6. System Requirements Specification

1. Initial Database Size

The initial database will be designed to support approximately 30 staff records, 5 manager records, and 3 admin records, along with 50 menu items, 100 ingredient records, 100 inventory records, and approximately 1,500 order and payment records. This initial size is sufficient for system deployment in a single restaurant environment and supports daily operational needs during the early stage of system usage.

2. Database Growth

The database is expected to grow continuously as the restaurant operates on a daily basis:

- Approximately 50–100 new order and payment records will be added per day.
- Menu data will grow slowly, with approximately 2–5 new menu items added per month.
- Ingredient and inventory records will be updated frequently due to stock usage and restocking, but the number of records will remain relatively stable.
- Staff, manager, and admin records will change occasionally, with approximately 1–2 staff records added or removed per month.

The system must support long-term data growth without requiring major structural changes to the database schema.

3. Record Search Types

The system must support frequent data searches during business hours, with the ability to handle up to 150 queries per minute during peak periods. Typical search operations include:

- Menu Searches: Staff and managers searching menu items by category, name, or price.
- Order Searches: Staff searching orders by order ID, order status, or date.
- Inventory Searches: Checking ingredient availability, current stock levels, and minimum stock thresholds.
- Staff and User Searches: Admins querying staff, manager, and admin records for system management purposes.

4. Networking and Shared Access

The system will operate over the restaurant's internal network and support multiple concurrent users, including admins, managers, and staff.

Shared database access will be managed to ensure data consistency, particularly during order processing and inventory updates.

Network access will be restricted to authorized devices and users within the restaurant environment.

5. Performance

The system should ensure that:

- The average database query response time is under 3 seconds for at least 95% of transactions.
- Order confirmation, payment recording, and inventory updates occur in near real time.
- The system supports at least 10 concurrent users without noticeable performance degradation.

Efficient indexing, query optimization, and proper database design techniques will be applied to meet performance requirements.

6. Security

To protect sensitive system data, the following security mechanisms will be implemented:

- Role-Based Access Control (RBAC): Access privileges will be defined based on user roles (Admin, Manager, and Staff), ensuring that users can only access data relevant to their responsibilities.
- Authentication: Admins, managers, and staff must log in before accessing system functions.
- Data Protection: Sensitive information such as citizen ID, login credentials, and contact details will be securely stored and protected from unauthorized access.
- Session Control: Automatic session timeout will be enforced after a period of inactivity to reduce security risks.

7. Backup and Recovery

- Full database backups will be performed once every 24 hours.
- Incremental backups will be performed every 4 hours.
- The system will allow database recovery with a maximum data loss of no more than 4 hours.
- Backup data will be stored securely and tested periodically to ensure successful recovery.

8. Legal Issues

The system will comply with applicable legal and ethical requirements:

- Personal Data Protection: Customer and staff data will be handled in accordance with data protection laws such as PDPA.
- Payment Data Handling: Payment-related information will be processed securely and not stored unnecessarily.
- Data Retention Policy: Customer and transaction data will be retained only as long as required for business and legal purposes, after which it will be securely deleted.

References

- [1] DB Schema Example, "Restaurant Database Schema," Scribd. [Online]. Available: <https://fr.scribd.com/document/455694826/DB-Schema-docx>. [Accessed: Jan. 28, 2026].
- [2] Database Sample, "Restaurant Database Sample," DatabaseSample.com. [Online]. Available: <https://databasesample.com/database/restaurant-database>. [Accessed: Jan. 28, 2026].